

Problema 1.

a) Demonstrati ca $\Delta r \approx -\frac{\epsilon g(r)}{f'(r)}$

Stiim ca r , radacina a lui $f(x) \Rightarrow f(r) = 0$

Cum $\Delta r + r$ solutie pentru $f(x) + \epsilon g(r) \Rightarrow f(\Delta r + r) + \epsilon g(\Delta r + r) = 0$ (1)

Aplicam pentru $f(r + \Delta t)$ polinomul Taylor de ordin 1 cu $a = r$

$f(r + \Delta t) = f(r) + \Delta r f'(r)$, dar cum stiim ca $f(r) = 0 \Rightarrow f(r + \Delta t) = \Delta r f'(r)$

Aplicam acelasi lucru si pentru $g(\Delta r + r)$, cu $a = r$

$g(\Delta r + r) = g(r) + \Delta r g'(r)$

Inlocuim $f(\Delta r + r)$ si $g(\Delta r + r)$ in (1) $\Rightarrow \Delta r \cdot f'(r) + \epsilon \cdot [g(r) + \Delta r \cdot g'(r)] \approx 0$

Cum termenul $\epsilon \cdot \Delta r \cdot g'(r)$ este foarte foarte mic, il vom neglija $\Rightarrow \Delta r \cdot f'(r) + \epsilon g(r) = 0 \Rightarrow \Delta r \approx -\frac{\epsilon g(r)}{f'(r)}$

b)

```
r_vechi = 6;
epsilon = -10^(-6);

g_r = r_vechi^7;
f_prim_r = prod(6 - (1:5));

delta_r = - (epsilon * g_r) / f_prim_r;
r_estimat_6 = r_vechi + delta_r;

fprintf('1. Estimarea radacinii de langa 6: %f\n', r_estimat_6);
```

1. Estimarea radacinii de langa 6: 6.002333

```
r_estimat_max = 1 / abs(epsilon);

fprintf('2. Estimarea celei mai mari radacini: %e\n\n', r_estimat_max);
```

2. Estimarea celei mai mari radacini: 1.000000e+06

```
coefs_f = poly(1:6);
```

```
P = [0, coefs_f];
P(1) = epsilon;
```

```
radacini_exacte = roots(P);
radacini_exacte = sort(radacini_exacte);
```

```
fprintf('Radacina exacta de langa 6 MATLAB: %f\n', radacini_exacte(6));
```

Radacina exacta de langa 6 MATLAB: 6.002327

```
fprintf('Cea mai mare radacina exacta MATLAB: %e\n', radacini_exacte(7));
```

Cea mai mare radacina exacta MATLAB: 9.999790e+05

Problema 2.

$$P(x) = x^{20} - (x-1)(x-2)(x-3)(x-4)(x-5)(x-6)$$

Conditionarea coeficientilor

Pentru un polinom $P(x) = \sum_{k=0}^n c_k x^k$, calculam numarul de conditionare relativ la o radacina dupa formula

$$K(r) = \frac{\sum_{k=1}^n |a_k| \cdot |r|^{n-k}}{|r| \cdot |P'(r)|}$$

```
polinom1 = zeros (1,21);
polinom1(1) = 1;

polinom2=poly(1:6);
polinom2 = [zeros(1,21 - length(polinom2)),polinom2];

P = polinom1 - polinom2;

radacini = roots(P);

for i = 1:length(radacini)
    xi = radacini(i);
    nr_cond = condpol(P, xi);

    fprintf('Radacina %2d: %9.4f + %9.4f | Cond: %e\n', ...
            i, real(xi), imag(xi), nr_cond);
end
```

Radacina 1:	-1.6000 +	0.0000	Cond: 5.601095e-02
Radacina 2:	-1.4955 +	0.5488	Cond: 6.046914e-02
Radacina 3:	-1.4955 +	-0.5488	Cond: 6.046914e-02
Radacina 4:	-1.2006 +	1.0151	Cond: 7.609759e-02
Radacina 5:	-1.2006 +	-1.0151	Cond: 7.609759e-02
Radacina 6:	-0.7666 +	1.3334	Cond: 1.115922e-01
Radacina 7:	-0.7666 +	-1.3334	Cond: 1.115922e-01
Radacina 8:	-0.2666 +	1.4677	Cond: 1.900908e-01
Radacina 9:	-0.2666 +	-1.4677	Cond: 1.900908e-01

Radacina 10:	0.2217 +	1.4179	Cond: 3.724191e-01
Radacina 11:	0.2217 +	-1.4179	Cond: 3.724191e-01
Radacina 12:	0.6334 +	1.2161	Cond: 8.213178e-01
Radacina 13:	0.6334 +	-1.2161	Cond: 8.213178e-01
Radacina 14:	0.9294 +	0.9143	Cond: 1.970093e+00
Radacina 15:	0.9294 +	-0.9143	Cond: 1.970093e+00
Radacina 16:	1.0991 +	0.5666	Cond: 4.975821e+00
Radacina 17:	1.0991 +	-0.5666	Cond: 4.975821e+00
Radacina 18:	1.1494 +	0.2073	Cond: 1.475255e+01
Radacina 19:	1.1494 +	-0.2073	Cond: 1.475255e+01
Radacina 20:	0.9929 +	0.0000	Cond: 3.549544e+01

```

m = 100;
dispersie = 1e-2;
sigma = sqrt(dispersie);

figure('Name', 'Experiment Perturbare', 'Color', 'w');
hold on;
grid on;

plot(real(radacini), imag(radacini), 'ro', 'MarkerSize', 8, 'LineWidth', 2,
'DisplayName', 'Radacini Exacte');

for j = 1:m

    delta = sigma * randn(size(P));

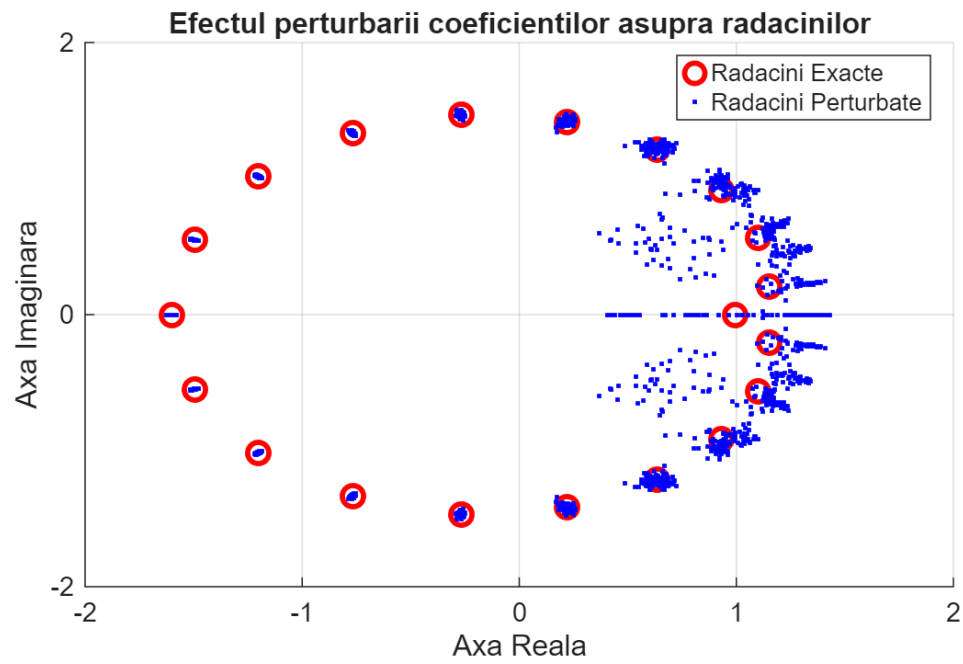
    P_perturbat = P .* (1 + delta);

    radacini_pert = roots(P_perturbat);

    if j == 1
        plot(real(radacini_pert), imag(radacini_pert), 'b.', 'MarkerSize', 5,
'DisplayName', 'Radacini Perturbate');
    else
        plot(real(radacini_pert), imag(radacini_pert), 'b.', 'MarkerSize', 5,
'HandleVisibility', 'off');
    end
end

title('Efectul perturbarii coeficientilor asupra radacinilor');
xlabel('Axa Reala');
ylabel('Axa Imaginara');
legend('Location', 'best');
hold off;

```



```
function nr_cond=condpol(coefs,xi)
    coefs_der=polyder(coefs);
    nr_cond=polyval(abs(coefs(2:end)),abs(xi))/abs(xi*polyval(coefs_der,xi));
end
```